CASE STUDY · DATA ANALYTICS

Olist E-Commerce Sales & Operations Analytics

End-to-end data analytics project using Python, PostgreSQL, SQL, Power BI and DAX.

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THE BUSINESS QUESTION

Reliable e-commerce performance starts with the right questions



One view

How can sales performance, order demand and fulfilment quality be monitored in one view?



Business contribution

Which categories contribute most to item-price sales, and how efficiently are orders delivered?



Signal vs noise

How can data distinguish real business movement from lifecycle, filter or data-quality issues?

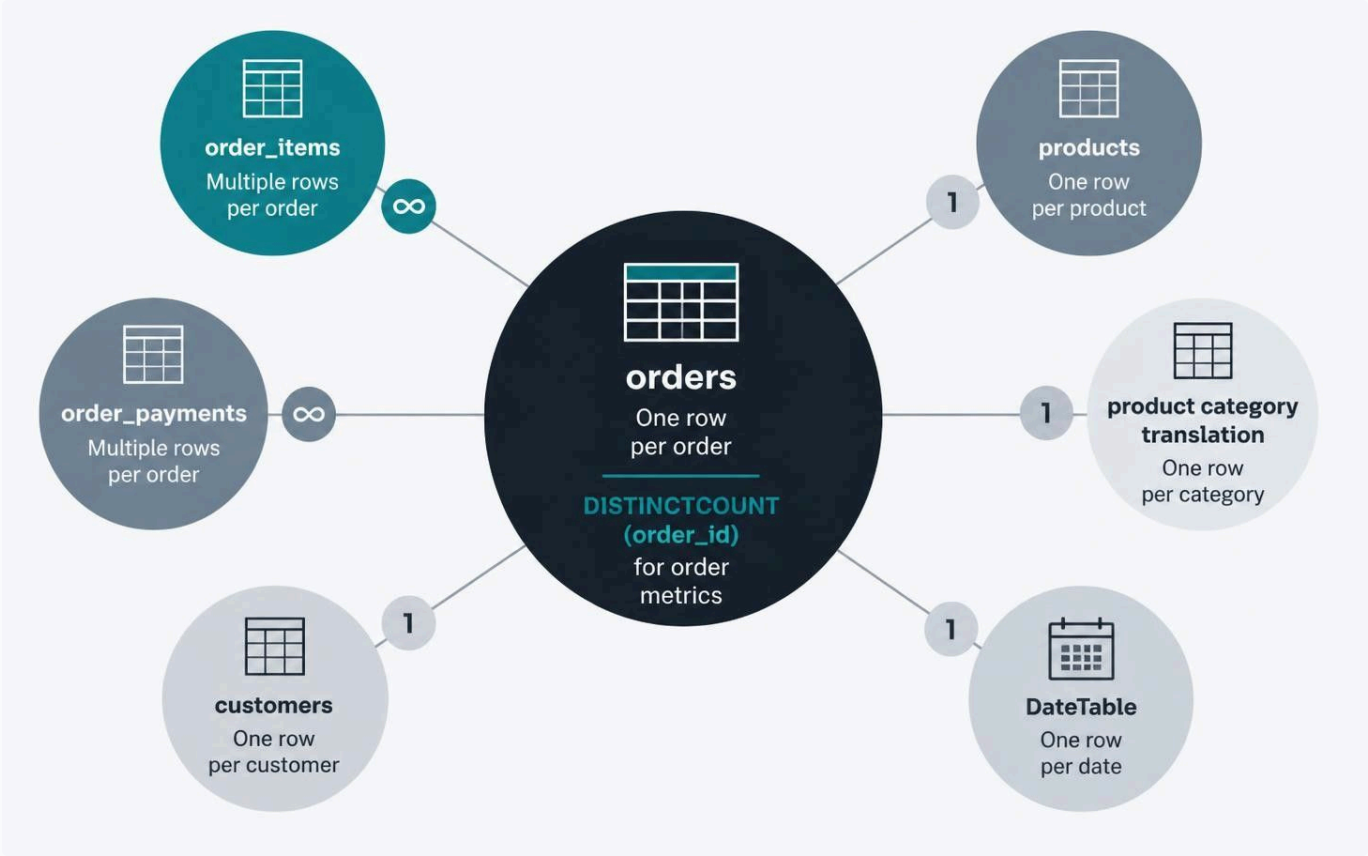
Objective: move from raw marketplace data to a reliable, interactive decision-support view.

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DATA FOUNDATION

Seven linked tables create one analytical foundation

Source: Brazilian E-Commerce Public Dataset by Olist on Kaggle.



Key control: use DISTINCTCOUNT(order_id) for order metrics to avoid multiplying orders across item or payment joins.

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END-TO-END WORKFLOW

Each tool owns a different stage of the analytical workflow

Each handoff makes the next layer more testable and decision-ready.

Kaggle → Python

Loading, cleaning and exploratory analysis.

PostgreSQL

Relational storage and SQL validation.

Power BI + DAX

Relationships, measures and interactive reporting.

Dashboard

Findings, recommendations and validation priorities.

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PYTHON PROFILING

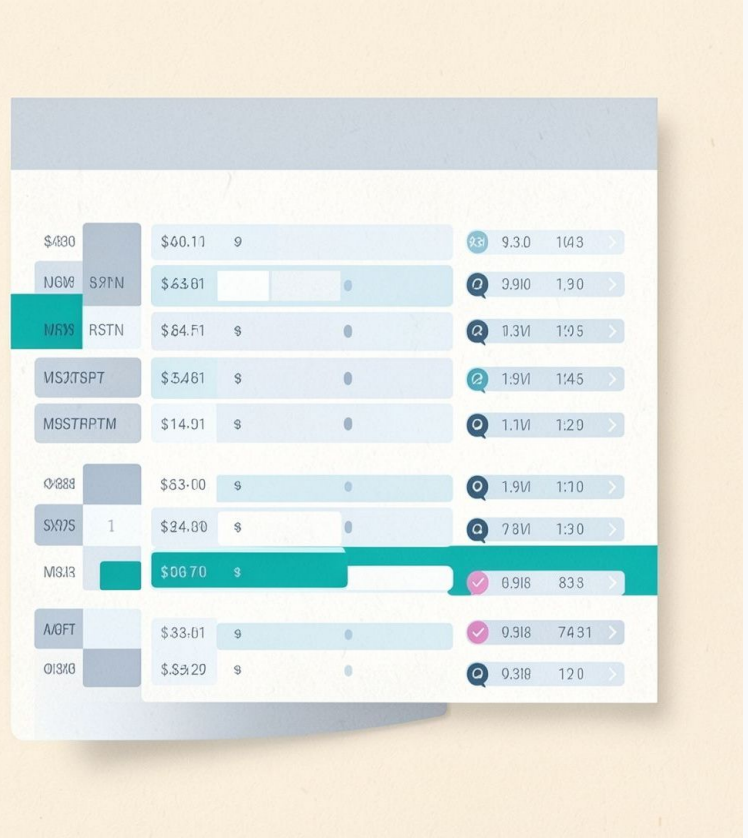
Python profiling made data quality visible before modelling

Initial orders: 99,441 rows, 8 columns; duplicate order IDs: 0.

Missing timestamps	Count
order_approved_at	160
order_delivered_carrier_date	1,783
order_delivered_customer_date	2,965

Converted timestamps, inspected status-specific nulls, calculated delivery duration and merged product/category attributes.

Context before deletion: Missing values were assessed in order-lifecycle context, not deleted blindly.



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POSTGRESQL · SQL

SQL turned the relational model into testable business measures

Database: **olist_analytics**

Queries validated counts, status distribution, monthly trends and category sales.

READABLE VALIDATION QUERY

```
SELECT order_status,
    COUNT(DISTINCT order_id) AS total_orders
FROM orders
GROUP BY order_status
ORDER BY total_orders DESC;
```

Techniques: COUNT, COUNT DISTINCT, GROUP BY, ORDER BY, DATE_TRUNC, JOIN, COALESCE, FILTER and EXTRACT(EPOCH).

Reporting risk exposed: an earlier monthly summary view ended before the latest raw-data timestamp.

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POWER BI · DAX

Power BI connects the model; DAX defines the business logic

Relationships

From	To
DateTable	orders
orders	order_items and order_payments
products	order_items
customers	orders
category translation	products

Core measures

Total Sales · Total Orders · Total Delivered Orders · On-Time Delivery Rate · Cancellation Rate · Average Delivery Time

Total Orders uses DISTINCTCOUNT(order_id).

Total Sales sums order_items[price] and is item-price sales, not profit or necessarily total customer spend.

Payment-type filtering requires validation because an order can have multiple payment records.

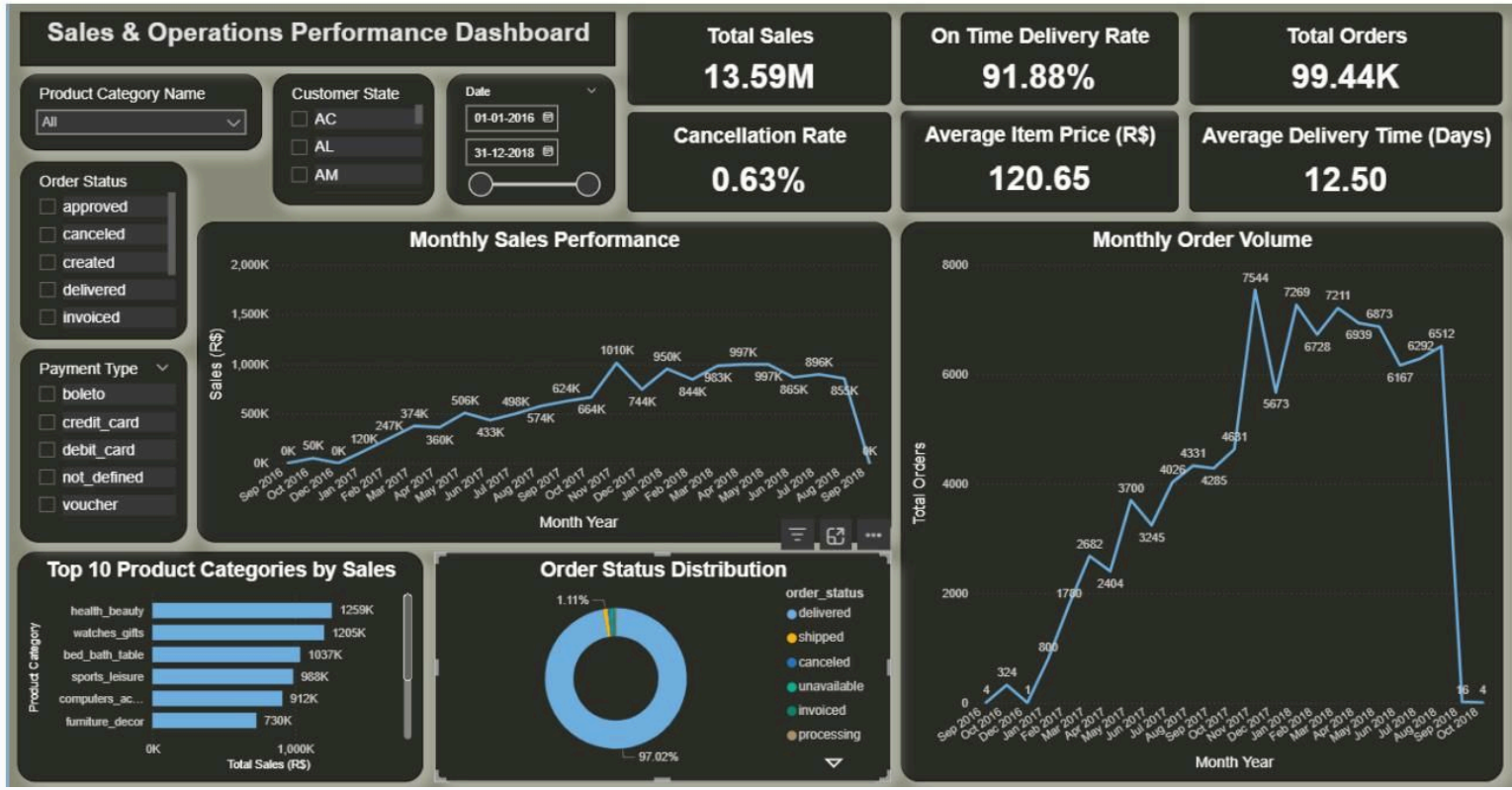
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EXECUTIVE VIEW

The dashboard turns the model into an executive monitoring view

KPIs, monthly sales and order trends, top categories, order-status composition and interactive slicers share one view.

Slicers: product category, customer state, date, order status and payment type.



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CURRENT DASHBOARD READOUT

Current dashboard KPIs show strong delivery performance with clear caveats

These are development-stage dashboard figures and require final reconciliation against SQL outputs.

Total Sales

R$13.59M

Sum of item prices, not profit or net revenue.

Total Orders

99.44K

Delivered Orders: 96.48K

Cancellation Rate

0.63%

On-Time Delivery Rate

91.88%

Average Delivery Time

12.50 days

Average Item Price

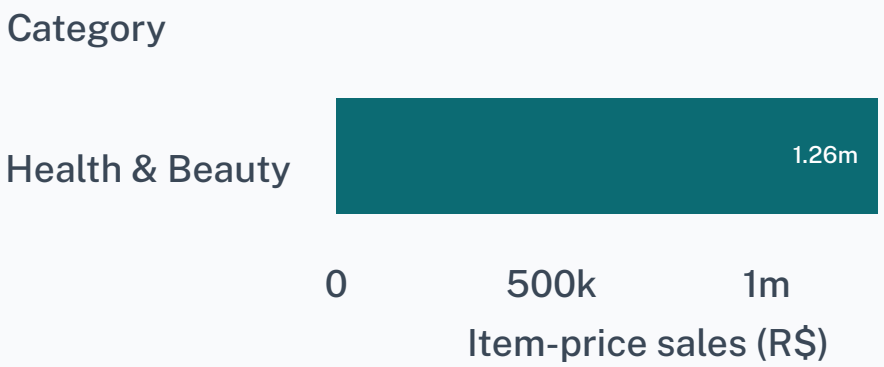
R$120.65

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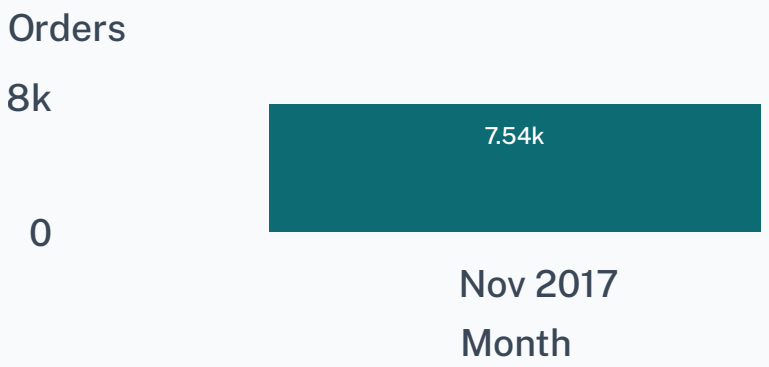
WHAT TO INVESTIGATE NEXT

Category mix and monthly demand point to focused business questions

Displayed category · item-price sales



Highest visible month · SQL order trend



Category questions

Health & Beauty leads the displayed categories by item-price sales at R$1,258,681.34. Watches & Gifts is another major contributor; Bed, Bath & Table has the highest displayed item count at 11,115.

Trend question

November 2017 is the highest visible month in the SQL order trend at 7,544 orders. The end-of-period decline is an open investigation, not a confirmed business event.

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